



Autumn Examinations 2014/2015

Exam Codes	3BA1, 3BPP2, 3BS9, 3MR3, 4BPT1, 4BS2 4BME1, 1HA1, 1HDS1, 1OA1, 1EM1
Exams	Third Arts, Third Science, Fourth Science, Fourth Mathematics and Education, Higher Diploma in Applied Science
Module	Applied Statistics I
Module Codes	ST311 MA5103
External Examiner	Professor S. T. Buckland
Internal Examiner(s)	Dr. E. Holian Professor J. Hinde
<u>Instructions:</u>	Answer any 3 questions from Questions 1-5 All Questions carry equal marks.
Duration	2 Hours
No. of Pages	22 pages, including this one
School	School of Mathematics, Statistics and Applied Mathematics
Release in Exam Venue	Yes ☐ No ☒
MCQ	Yes ☐ No ☒
<u>Requirements:</u>	
Statistical Tables/ Log Tables	Relevant distribution tables are attached to this paper. The <i>New Formulae & Tables</i> are optional.
Other Materials Allowed	A calculator is allowed (non-programmable and not capable of storing text).

Question 1

Complete parts (a), (b) and (c).

- (a) Cuckoos lay their eggs in the nests of host species. A researcher set out to assess whether cuckoo eggs can be distinguished from the host eggs by their size. A sample of cuckoo eggs were observed to have the following lengths, recorded in *mm*: 12, 15.5, 17, 18, 20.4, 20.7, 21, 21.5, 21.5, 21.7.

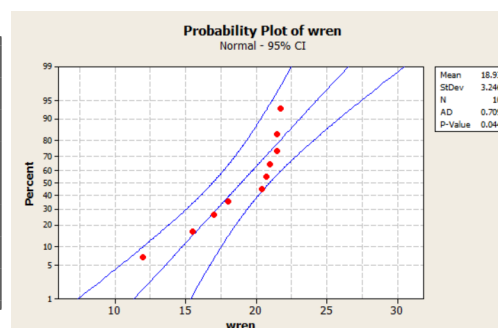
A summary of descriptive statistics and plots for this sample is provided below:

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
length	10	0	18.93	1.03	3.25	12.00	16.63	20.55	21.50	21.70

N for				
Variable	Mode	Mode	Skewness	Kurtosis
length	21.5	2	-1.25	0.85

Letter Value Display: length

	Depth	Lower	Upper	Mid	Spread
N=	10				
M	5.5	20.550	20.550	20.550	
H	3.0	17.000	21.500	19.250	4.500
E	2.0	15.500	21.500	18.500	6.000
	1	12.000	21.700	16.850	9.700



- (i) By comparing the values of the mean, the median and the mode(s) for this sample what does this tell us about the shape of the distribution of the sample? [2 marks]
- (ii) Show how the letter values are calculated for this sample. [4 marks]
- (iii) For what purpose are the letter values used, and what do these particular letter values indicate? [4 marks]
- (iv) What does the normal probability plot suggest? [2 marks]
- (b) The researcher wishes to determine whether this sample gives evidence that the typical length for the population of all cuckoo eggs is less than 21mm.
- (i) It is suggested that a 1-sample t-test be carried out to infer whether the average length for the population of all cuckoo eggs is less than 21mm. What can you conclude as a result of this analysis? Reference the output to justify your answer. [3 marks]

One-Sample T: length

Test of $\mu = 21$ vs < 21

Variable	N	Mean	StDev	SE Mean	95% Upper Bound	T	P
length	10	18.93	3.25	1.03	20.81	-2.02	0.037

Question 1 continues on the next page

Question 1 continued...

- (ii) Do you have any concerns about applying the 1-sample t-test in this application? If so, explain reasons for your concern. [2 marks]
- (iii) It is suggested that a 1-sample Signs test be carried out to infer whether the median length for the population of all cuckoo eggs is less than 21mm.

Sign Test for Median: length

Sign test of median = 21.00 versus < 21.00

	N	Below	Equal	Above	P	Median
length	10	6	1	3	0.254	20.55

Verify, by calculation, the value of the p-value as printed in the output for the Signs test. Include explanation of the formulation of the p-value. [4 marks]

- (iv) If testing at $\alpha = 0.05$, what can you conclude as a result of the Signs test? Include a statement of the hypotheses being tested, your decision and conclusion in this application. [3 marks]

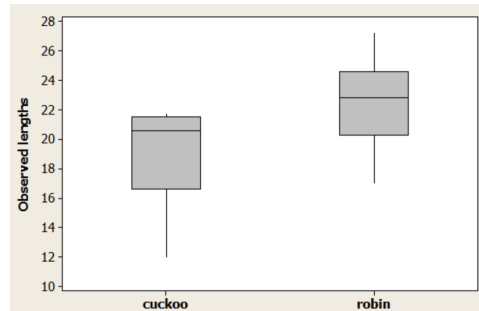
- (c) In addition to the observing the lengths of the sample of cuckoo eggs, the lengths of a sample of robins' eggs were recorded. The observed lengths for both samples are listed below with some summary statistics and plots provided.

Cuckoo 12, 15.5, 17, 18, 20.4, 20.7, 21, 21.5, 21.5, 21.7.

Robin 17, 20, 20.5, 21.3, 22.8, 24.2, 24.2, 25, 27.2.

Descriptive Statistics:

Variable	N	Mean	StDev	Median
cuckoo	10	18.93	3.25	20.55
robin	9	22.47	3.09	22.80



The researcher wishes to compare the typical length of cuckoos eggs to the typical length of eggs laid by the robin, in particular whether the cuckoo lengths are less than robins' eggs.

- (i) It is suggested to you to carry out a Wilcoxon Rank Sum test. Explain why, in this application, this test is more suitable than the 2-sample Z test, or the 2-sample t test. [4 marks]
- (ii) Carry out the test at significance level $\alpha=0.05$. Include a statement of the hypotheses being tested, formulation of the test statistic, the rejection region for the test statistic, and indicate clearly your decision and conclusion from the result of the test. [7 marks]

[Total 35 marks]

Question 2

Complete parts (a), (b) and (c).

- (a) (i) "The true correlation coefficient between two variables X and Y for a population of individuals is known to have value $\rho = 0$. This implies that X and Y are unrelated."
- Is this statement true or false? If you deem the statement to be false give a reason for your answer. [2 marks]
- (ii) A sample of 20 individuals gave a sample correlation coefficient of $r = 0.55$. Carry out a hypothesis test of whether there is evidence of a significant increasing relationship between X and Y in the population. Clearly state the null and alternative hypotheses of your test, and give a test statistic, a rejection region, your decision and your conclusion. [6 marks]

(b) Consider a simple linear regression model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$.

- (i) What does the error term, ϵ_i , represent? [2 marks]
- (ii) When carrying out an Analysis of Variance, explain in general what the terms SS_{Total} , SS_{Reg} and SS_{Error} represent. [5 marks]

(c) The following model is being considered in order to model the variability in a response variable Y , using five candidate predictor variables, X_1, X_2, X_3, X_4, X_5 ,

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \beta_5 X_{5i} + \epsilon_i.$$

The model is fitted to observations obtained for a sample of 20 individuals, and the results of the fit are provided in the partial printout on the next page.

Use the output to answer the following questions.

- (i) Calculate the coefficient of determination R^2 for this regression fit. Interpret the meaning of this value. [2 marks]
- (ii) Find the missing values in the ANOVA table, and complete the test, at the $\alpha = 0.05$ level, including statements of the hypotheses being tested, a rejection region, a decision and conclusion. (*You do not have to calculate the p-value.*) [6 marks]
- (iii) Complete the missing t-test in the output, i.e. test whether X_4 is a good predictor of Y at the $\alpha = 0.05$ level. Include the hypotheses being tested, the calculation of the test statistic, a rejection region, a decision and conclusion. [5 marks]
- (iv) Use the partial F-test to test the hypothesis : $H_0 : \beta_4 = \beta_5 = 0$ at $\alpha = 0.05$. Interpret your findings. [5 marks]
- (v) The output provides a prediction interval and a confidence interval for the response when $X_1 = 20$, $X_2 = 8$, $X_3 = 10$, $X_4 = 45$, $X_5 = 15$. Explain the difference as to when a prediction interval and a confidence interval applies. [2 marks]

Question 2 continues on the next page

Question 2 continued...

The regression equation is

$$Y = 110 + 2.74 X_1 + 6.55 X_2 - 4.37 X_3 + 2.15 X_4 - 1.27 X_5$$

Predictor	Coef	SE Coef	T	P
Constant	110.00	38.190	2.87	0.012
X1	2.74	0.748	3.67	0.003
X2	6.55	2.560	2.56	0.023
X3	-4.37	3.626	-1.21	0.248
X4	2.15	0.363	-	-
X5	-1.27	0.903	-1.40	0.182

S = 14.1212 R-Sq = - % R-Sq(adj) = - %

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	-	15526.0	-	-	-
Residual Error	-	2791.7	-		
Total	19	18317.8			

Source	DF	Seq SS
X1	1	5547.6
X2	1	661.1
X3	1	256.2
X4	1	8668.3
X5	1	392.8

Predicted Values for New Observations

New					
Obs	Fit	SE Fit	95% CI	95% PI	
1	250.81	3.56	(243.17, 258.45)	(219.57, 282.05)	

Values of Predictors for New Observations

New					
Obs	X1	X2	X3	X4	X5
1	20.0	8.00	10.0	45.0	15.0

[Total 35 marks]

Question 3

The manager at a large clothing retailer is interested in predicting how much customers will spend on their next purchase. Data for a random sample of 25 customers were collected including the amount spent in the customer's latest purchase, *Amount*, and six possible explanatory variables:

- *Recency*, the number of months since the customer's last purchase
- *Freq12*, number of purchases in the last 12 months
- *Total12*, total amount spent over all transactions the customer made with the retailer in last 12 months.
- *Freq24*, number of purchases in the last 24 months
- *Total24*, total amount spent over all transactions the customer made with the retailer in last 24 months.
- *Card*, value 1 if the customer had a store card, value 0 otherwise.

- (a) A multiple linear regression model using three of the explanatory variables, *Freq12*, *Total12* and *Card*, was first considered giving the following output:

The regression equation is
 $\text{Amount} = 158 - 38.9 \text{ Freq12} + 0.287 \text{ Total12} + 41.0 \text{ Card}$

Predictor	Coef	SE Coef	T	P
Constant	158.09	28.02	5.64	0.000
Freq12	-38.939	5.647	-6.90	0.000
Total12	0.28665	0.04140	6.92	0.000
Card	41.03	23.09	1.78	0.090

S = 53.1833 R-Sq = 83.4% R-Sq(adj) = 81.0%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	3	298320	99440	35.16	0.000
Residual Error	21	59398	2828		
Total	24	357717			

- (i) Interpret the meaning of the estimated coefficients of the three predictor variables *Freq12*, *Total12* and *Card*. [5 marks]
- (ii) Write down the hypotheses, in terms of the model coefficients, β 's, being tested in the ANOVA table for this model.

What do you conclude as a result of this test:

- in terms of the model coefficients, β 's, and
- in terms of the predictor variables?

[5 marks]

Question 3 continues on the next page

Question 3 continued...

- (iii) Comment on the significance of the three predictor variables *Freq12*, *Total12* and *Card*. Use the output to support your answer. Include a statement of the hypotheses being tested for any hypothesis test you reference.

[6 marks]

- (b) A model with just the three predictor variables *Recency*, *Freq12* and *Freq24* is fitted to the data giving the following output:

The regression equation is					
Amount = 197 + 12.3 Recency - 15.0 Freq12 - 6.3 Freq24					
Predictor	Coef	SE Coef	T	P	VIF
Constant	197.2	114.5	1.72	0.100	
Recency	12.30	11.56	1.06	0.300	1.952
Freq12	-15.02	18.58	-0.81	0.428	2.710
Freq24	-6.31	12.17	-0.52	0.610	2.104
S = 107.453 R-Sq = 32.2% R-Sq(adj) = 22.5%					
Analysis of Variance					
Source	DF	SS	MS	F	P
Regression	3	115249	38416	3.33	0.039
Residual Error	21	242469	11546		
Total	24	357717			
Source	DF	Seq SS			
Recency	1	92659			
Freq12	1	19488			
Freq24	1	3102			

Discuss the predictive power of the three variables, by giving your conclusions from the outputs of the t-tests and ANOVA test provided. Ensure you provide the hypotheses being tested for any test that you reference.

[8 marks]

- (c) The following output was produced using the best subsets routine in which five of the six predictors are considered.

- (i) Explain the meaning of the phrase “a parsimonious model”. [2 marks]

- (ii) Write down the most suitable model, i.e. the set of predictors, that you think best explains the variability in observed **Amount**.

Justify your choice of model by explaining the role of s , R^2 , adjusted R^2 , and Mallow's C_p in choosing the best model. [9 marks]

Question 3 continues on the next page

Question 3 continued...

Response is Amount									
					R T T				
					e F o F o				
					c r t r t				
					e e a e a				
					n q l q l				
					c 1 1 2 2				
					y 2 2 4 4				
Vars	R-Sq	R-Sq(adj)	Cp	S					
1	42.6	40.1	53.1	94.495		X			
1	31.6	28.7	67.2	103.11				X	
2	80.9	79.2	5.6	55.730		X	X		
2	74.9	72.6	13.4	63.918	X	X			
3	84.8	82.6	2.7	50.937	X	X	X		
3	81.0	78.2	7.6	56.960		X	X	X	
4	85.2	82.3	4.0	51.370	X	X	X	X	
4	84.9	81.9	4.5	51.936	X	X	X	X	
5	85.3	81.4	6.0	52.649	X	X	X	X	X

[Total 35 marks]

Question 4

Complete parts (a), (b) and (c).

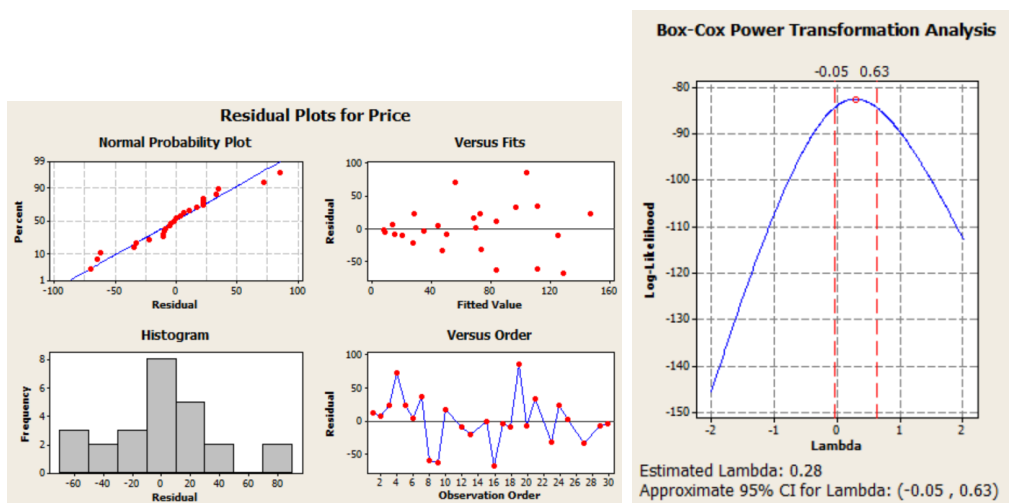
- (a) When fitting a simple linear regression model to data what assumptions are required? [5 marks]
- (b) The output and diagnostic plots provided below show the results of a simple linear regression fit to the price, *Price*, observed for a sample of 25 second-hand textbooks, using number of pages, *Pages*, as a predictor.

The regression equation is

$$\text{Price} = 1.1 + 0.138 \text{ Pages}$$

Predictor	Coef	SE Coef	T	P
Constant	1.12	14.27	0.08	0.938
Pages	0.138	0.02595	5.32	0.000

S = 37.7987 R-Sq = 55.2% R-Sq(adj) = 53.2%



- (i) Comment on the validity of the assumptions for this regression fit. [5 marks]
- (ii) What does the output of the Box-Cox transformation indicate? [5 marks]
- (iii) A simple linear regression using *Pages* as the predictor was fitted to the transformed response variable $\ln(\text{Price})$ giving the output below:

The regression equation is

$$\ln(\text{Price}) = 2.28 + 0.003 \text{ Pages}$$

Predictor	Coef	SE Coef	T	P
Constant	2.2763	0.2834	8.03	0.000
Pages	0.003	0.0005153	5.87	0.000

S = 0.750440 R-Sq = 60.0% R-Sq(adj) = 58.2%

Question 4 continues on the next page

Question 4 continued...

Interpret the meaning of the estimated coefficient of *Pages* in this model.

[5 marks]

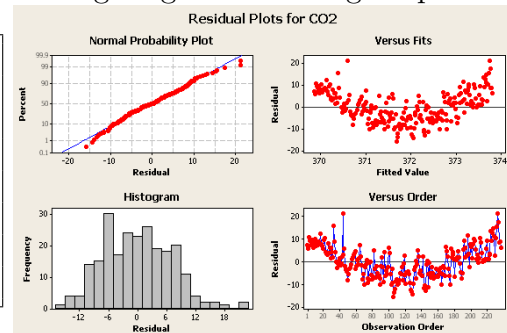
(iv) Using the model in part (iii) predict the price of a book which has 150 pages.

[4 marks]

- (c) Scientists at a research station in Germany wish to build a model to explain variability in CO₂ levels in the atmosphere. They recorded the CO₂ levels, measured in parts per million, for each day from the start of April 2001 through to November 2001, and used time as a possible explanatory variable, defined as day of the year, i.e. observing CO₂ for days 91-334.

A simple linear regression is fitted to the sample data giving the following output:

The regression equation is CO ₂ = 368 + 0.0162 Day				
Predictor	Coef	SE Coef	T	P
Constant	368	1.451	253.96	0.000
Day	0.0162	0.006425	2.53	0.012
S = 6.96323 R-Sq = 2.6% R-Sq(adj) = 2.2%				



- (i) Comment on the validity of the assumptions using the diagnostic plots above.

[4 marks]

- (ii) A linear regression using Day and DaySq, calculated as Day², as predictors is fitted to the response variable CO₂ giving the following output:

The regression equation is CO ₂ = 415 - 0.476 Day + 0.00116 DaySq				
Predictor	Coef	SE Coef	T	P
Constant	415	2.856	145.28	0.000
Day	-0.476	0.02874	-16.57	0.000
DaySq	0.00116	0.00006684	17.32	0.000
S = 4.61913 R-Sq = 57.3% R-Sq(adj) = 57.0%				

Comment on the significance of the term DaySq.

[4 marks]

- (iii) Using the model in part (ii) predict the CO₂ in the atmosphere for day 200, i.e. mid July.

[3 marks]

[Total 35 marks]

Question 5

Complete parts (a), (b) and (c).

- (a) A used car salesman is interested in modelling the variability in price of cars. He has collected sample data from an internet site for three types of sports car, Porsche, BMW and Jaguar, recording their advertised price (in thousands of euro). To investigate if the different car types have an effect on price, indicator variables are defined for the three car types:

$$\begin{aligned} \text{IndP} : & \begin{cases} 1 & \text{if the price observed is for a used Porsche car} \\ 0 & \text{otherwise} \end{cases} \\ \text{IndB} : & \begin{cases} 1 & \text{if the price observed is for a used BMW car} \\ 0 & \text{otherwise} \end{cases} \\ \text{IndJ} : & \begin{cases} 1 & \text{if the price observed is for a used Jaguar car} \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

and the model, $\text{Price}_i = \beta_0 + \beta_1 \text{IndP}_i + \beta_2 \text{IndB}_i + \epsilon_i$, is fitted to the data, giving:

The regression equation is

$$\text{Price} = 32.0 + 18.6 \text{ IndP} - 1.72 \text{ IndB}$$

Predictor	Coef	SE Coef	T	P
Constant	31.957	2.775	11.52	0.000
IndP	18.580	3.924	4.73	0.000
IndB	-1.723	3.924	-0.44	0.662

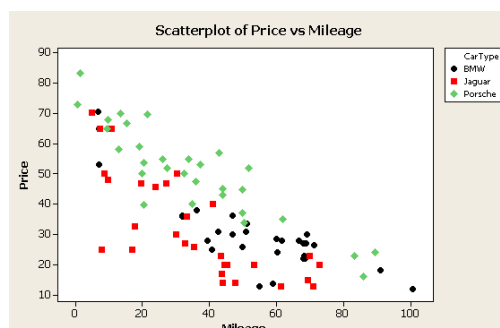
S = 15.1975 R-Sq = 27.5% R-Sq(adj) = 25.8%

- i) Use the fitted model to predict the price for:
- a Porsche car.
 - a Jaguar car.
 - a BMW car.
- [3 marks]
- ii) Interpret the meaning of the estimated coefficients of IndP and IndB , and comment on their significance. Ensure you state the hypotheses being tested when referring to the results of the output. [6 marks]
- iii) Write down an alternative model proposal that you would recommend be fitted to the observed data that would help assess the comparison of price between each of the car types, Jaguar and BMW, with a Porsche car. [2 marks]

Question 5 continues on the next page

Question 5 continued...

- (b) The salesman has also recorded the mileage (in thousands of miles) for the sample of cars observed. A plot of the observed relationship of price with mileage is provided, with an indication to the different car types.



The regression analysis below models the response, *Price*, by the predictor *Mileage*, and by the indicator variables *IndP*, *IndB*, as defined above.

The regression equation is

$$\text{Price} = 52.3 - 0.565 \text{ Mileage} + 18.0 \text{ IndP} + 8.05 \text{ IndB}$$

Predictor	Coef	SE Coef	T	P
Constant	52.300	2.172	24.06	0.000
Mileage	-0.565	0.04166	-13.57	0.000
IndP	18.000	2.227	8.08	0.000
IndB	8.050	2.340	3.44	0.001

S = 8.62451 R-Sq = 76.9% R-Sq(adj) = 76.1%

- Interpret the meaning of the fitted coefficients of *Mileage*, *IndP* and *IndB*, in this model. [5 marks]
 - For this model, write down the estimated relationship between price and mileage for each of the three car types, Porsche, BMW and Jaguar. [3 marks]
 - What assumption is being made about the relationship of price and mileage for the three different car types in this model? [2 marks]
- (c) Two interaction terms were added to the model:
Mileage * *P*, calculated as $(\text{Mileage}_i) \times (\text{IndP}_i)$ for each individual,
Mileage * *B*, calculated as $(\text{Mileage}_i) \times (\text{IndB}_i)$ for each individual,
giving the following output:

Question 5 continues on the next page

Question 5 continued...

The regression equation is					
Price = 54.2 - 0.620 Mileage + 16.9 IndP + 2.06 IndB + 0.031 Mileage*P + 0.130 Mileage*B					
Predictor	Coef	SE Coef	T	P	
Constant	54.2	3.186	17.02	0.000	
Mileage	-0.620	0.077	-8.05	0.000	
IndP	16.9	4.278	3.94	0.000	
IndB	2.06	5.236	0.39	0.695	
Mileage*P	0.031	0.1030	0.30	0.765	
Mileage*B	0.130	0.1057	1.23	0.221	
S = 8.63844 R-Sq = 77.4% R-Sq(adj) = 76.0%					
Analysis of Variance					
Source	DF	SS	MS	F	P
Regression	5	21429.8	4286.0	57.43	0.000
Residual Error	84	6268.3	74.6		
Total	89	27698.1			

- Interpret the meaning of the fitted coefficients of *Mileage * P* and *Mileage * B* in this model. [4 marks]
- For this model, write down the estimated relationship between price and mileage for each of the three car types, Porsche, BMW and Jaguar. [4 marks]
- Determine if the terms *Mileage * P* and *Mileage * B* are significant, using the results of the individual t-tests in this analysis. Ensure you state the hypotheses being tested when referring to the results of the output. [6 marks]

[Total 35 marks]

ST311 Formulae

Some useful z-scores:

$$Z_{0.1} = 1.28$$

$$Z_{0.05} = 1.64$$

$$Z_{0.025} = 1.96$$

$$Z_{0.01} = 2.33$$

$$Z_{0.005} = 2.58$$

Some ranges for the standard normal distribution:

$$1.35 \quad \text{middle } 50\% : Z_{0.75} - Z_{0.25}$$

$$2.30 \quad \text{middle } 75\% : Z_{0.875} - Z_{0.125}$$

$$3.07 \quad \text{middle } 87.5\% : Z_{0.9375} - Z_{0.0625}$$

$$3.73 \quad \text{middle } 93.75\% : Z_{0.96875} - Z_{0.03125}$$

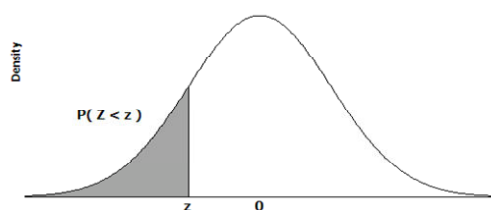
$$4.31 \quad \text{middle } 96.875\% : Z_{0.984375} - Z_{0.015625}$$

$$4.84 \quad \text{middle } 98.4375\% : Z_{0.9921875} - Z_{0.0078125}$$

Some formulae that may be useful when carrying out inference with parametric procedures:

- Inference for population correlation coefficient ρ : $TS = r\sqrt{\frac{n-2}{1-r^2}}$
- Partial F-test : $TS = \frac{(\text{additional SSreg})/q}{MS_{error}}$

Table: Cumulative Probabilities for the Standard Normal Distribution, $Z \sim \text{Normal}(0,1)$.



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Table: Cumulative Probabilities for the Standard Normal Distribution, $Z \sim \text{Normal}(0,1)$.

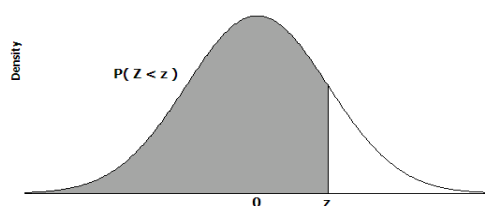
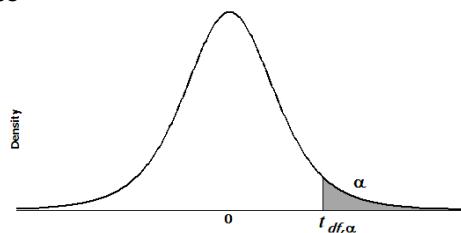
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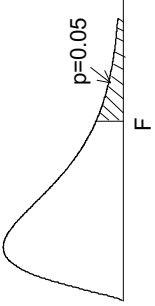
Table: t distribution critical values



df	Upper tail probability											
	0.25	0.20	0.15	0.10	0.05	0.025	0.02	0.01	0.005	0.0025	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.706	15.895	31.821	63.657	127.321	318.309	636.619
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.089	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	0.679	0.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	0.679	0.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
Z*	0.674	0.842	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.090	3.291

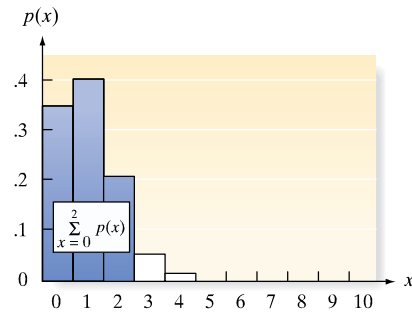
F distribution for $p = 0.05$

For various pairs of degrees of freedom (v_1, v_2), the tabled entries represent the critical values of F above which the proportion $p = 0.05$ of the distribution falls. (Example, for $df = 3, 8$ and $F = 2.92$ is exceeded by $p = 0.05$ of the population.



$p = 0.05$		Degrees of freedom for the numerator v_1																		
v_2	1	2	3	4	5	6	7	8	9	10	12	15	20	30	40	60	120	∞		
1	161.446	199.499	215.707	224.583	230.160	233.988	236.767	238.884	240.543	241.882	243.905	245.949	248.016	250.096	251.144	252.196	253.254	254.317		
2	18.513	19.000	19.164	19.247	19.296	19.329	19.353	19.371	19.385	19.396	19.412	19.429	19.446	19.463	19.471	19.479	19.487	19.496		
3	10.128	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.785	8.745	8.703	8.660	8.617	8.594	8.572	8.549	8.526		
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.912	5.858	5.803	5.746	5.717	5.688	5.658	5.628		
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735	4.678	4.619	4.558	4.496	4.464	4.431	4.398	4.365		
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	4.000	3.938	3.874	3.808	3.774	3.740	3.705	3.669		
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.575	3.511	3.445	3.376	3.340	3.304	3.267	3.230		
8	5.318	4.459	4.066	3.838	3.688	3.581	3.500	3.438	3.388	3.347	3.284	3.218	3.150	3.079	3.043	3.005	2.967	2.928		
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.073	3.006	2.936	2.864	2.826	2.787	2.748	2.707		
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.913	2.845	2.774	2.700	2.661	2.621	2.580	2.538		
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854	2.788	2.719	2.646	2.570	2.531	2.490	2.448	2.404		
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.687	2.617	2.544	2.466	2.426	2.384	2.341	2.296		
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.604	2.533	2.459	2.380	2.339	2.297	2.252	2.206		
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.534	2.463	2.388	2.308	2.266	2.223	2.178	2.131		
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.475	2.403	2.328	2.247	2.204	2.160	2.114	2.066		
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.425	2.352	2.276	2.194	2.151	2.106	2.059	2.010		
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.381	2.308	2.230	2.148	2.104	2.058	2.011	1.960		
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.342	2.269	2.191	2.107	2.063	2.017	1.968	1.917		
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.308	2.234	2.155	2.071	2.026	1.980	1.930	1.878		
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.278	2.203	2.124	2.039	1.994	1.946	1.896	1.843		
21	4.325	3.467	3.072	2.840	2.685	2.573	2.488	2.420	2.366	2.321	2.250	2.176	2.096	2.010	1.965	1.916	1.866	1.812		
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297	2.226	2.151	2.071	1.984	1.938	1.889	1.838	1.783		
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275	2.204	2.128	2.048	1.961	1.914	1.865	1.813	1.757		
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255	2.183	2.108	2.027	1.939	1.892	1.842	1.790	1.733		
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165	2.092	2.015	1.932	1.841	1.792	1.740	1.683	1.622		
40	4.085	3.232	2.839	2.606	2.449	2.336	2.249	2.180	2.124	2.077	2.003	1.924	1.839	1.744	1.693	1.637	1.577	1.509		
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	2.040	1.993	1.917	1.836	1.748	1.649	1.594	1.534	1.467	1.399		
120	3.920	3.072	2.680	2.447	2.290	2.175	2.087	2.016	1.959	1.910	1.834	1.750	1.659	1.554	1.495	1.429	1.352	1.254		
∞	3.841	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.880	1.831	1.752	1.666	1.571	1.459	1.394	1.318	1.221	1.000		

TABLE II Binomial Probabilities



Tabulated values are $\sum_{x=0}^k p(x)$. (Computations are rounded at the third decimal place.)

a. $n = 5$

$\begin{matrix} p \\ k \end{matrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.951	.774	.590	.328	.168	.078	.031	.010	.002	.000	.000	.000	.000
1	.999	.977	.919	.737	.528	.337	.188	.087	.031	.007	.000	.000	.000
2	1.000	.999	.991	.942	.837	.683	.500	.317	.163	.058	.009	.001	.000
3	1.000	1.000	1.000	.993	.969	.913	.812	.663	.472	.263	.081	.023	.001
4	1.000	1.000	1.000	1.000	.998	.990	.969	.922	.832	.672	.410	.226	.049

b. $n = 6$

$\begin{matrix} p \\ k \end{matrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.941	.735	.531	.262	.118	.047	.016	.004	.001	.000	.000	.000	.000
1	.999	.967	.886	.655	.420	.233	.109	.041	.011	.002	.000	.000	.000
2	1.000	.998	.984	.901	.744	.544	.344	.179	.070	.017	.001	.000	.000
3	1.000	1.000	.999	.983	.930	.821	.656	.456	.256	.099	.016	.002	.000
4	1.000	1.000	1.000	.998	.989	.959	.891	.767	.580	.345	.114	.033	.001
5	1.000	1.000	1.000	1.000	.999	.996	.984	.953	.882	.738	.469	.265	.059

c. $n = 7$

$\begin{matrix} p \\ k \end{matrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.932	.698	.478	.210	.082	.028	.008	.002	.000	.000	.000	.000	.000
1	.998	.956	.850	.577	.329	.159	.063	.019	.004	.000	.000	.000	.000
2	1.000	.996	.974	.852	.647	.420	.227	.096	.029	.005	.000	.000	.000
3	1.000	1.000	.997	.967	.874	.710	.500	.290	.126	.033	.003	.000	.000
4	1.000	1.000	1.000	.995	.971	.904	.773	.580	.353	.148	.026	.004	.000
5	1.000	1.000	1.000	1.000	.996	.981	.937	.841	.671	.423	.150	.044	.002
6	1.000	1.000	1.000	1.000	1.000	.998	.992	.972	.918	.790	.522	.302	.068

(continued)

TABLE II (continued)

d. $n = 8$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.923	.663	.430	.168	.058	.017	.004	.001	.000	.000	.000	.000	.000
1	.997	.943	.813	.503	.255	.106	.035	.009	.001	.000	.000	.000	.000
2	1.000	.994	.962	.797	.552	.315	.145	.050	.011	.001	.000	.000	.000
3	1.000	1.000	.995	.944	.806	.594	.363	.174	.058	.010	.000	.000	.000
4	1.000	1.000	1.000	.990	.942	.826	.637	.406	.194	.056	.005	.000	.000
5	1.000	1.000	1.000	.999	.989	.950	.855	.685	.448	.203	.038	.006	.000
6	1.000	1.000	1.000	1.000	.999	.991	.965	.894	.745	.497	.187	.057	.003
7	1.000	1.000	1.000	1.000	1.000	.999	.996	.983	.942	.832	.570	.337	.077

e. $n = 9$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.914	.630	.387	.134	.040	.010	.002	.000	.000	.000	.000	.000	.000
1	.997	.929	.775	.436	.196	.071	.020	.004	.000	.000	.000	.000	.000
2	1.000	.992	.947	.738	.463	.232	.090	.025	.004	.000	.000	.000	.000
3	1.000	.999	.992	.914	.730	.483	.254	.099	.025	.003	.000	.000	.000
4	1.000	1.000	.999	.980	.901	.733	.500	.267	.099	.020	.001	.000	.000
5	1.000	1.000	1.000	.997	.975	.901	.746	.517	.270	.086	.008	.001	.000
6	1.000	1.000	1.000	1.000	.996	.975	.910	.768	.537	.262	.053	.008	.000
7	1.000	1.000	1.000	1.000	1.000	.996	.980	.929	.804	.564	.225	.071	.003
8	1.000	1.000	1.000	1.000	1.000	1.000	.998	.990	.960	.866	.613	.370	.086

f. $n = 10$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.904	.599	.349	.107	.028	.006	.001	.000	.000	.000	.000	.000	.000
1	.996	.914	.736	.376	.149	.046	.011	.002	.000	.000	.000	.000	.000
2	1.000	.988	.930	.678	.383	.167	.055	.012	.002	.000	.000	.000	.000
3	1.000	.999	.987	.879	.650	.382	.172	.055	.011	.001	.000	.000	.000
4	1.000	1.000	.998	.967	.850	.633	.377	.166	.047	.006	.000	.000	.000
5	1.000	1.000	1.000	.994	.953	.834	.623	.367	.150	.033	.002	.000	.000
6	1.000	1.000	1.000	.999	.989	.945	.828	.618	.350	.121	.013	.001	.000
7	1.000	1.000	1.000	1.000	.998	.988	.945	.833	.617	.322	.070	.012	.000
8	1.000	1.000	1.000	1.000	1.000	.998	.989	.954	.851	.624	.264	.086	.004
9	1.000	1.000	1.000	1.000	1.000	1.000	.999	.994	.972	.893	.651	.401	.096

(continued)

TABLE II (continued)

g. $n = 15$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.860	.463	.206	.035	.005	.000	.000	.000	.000	.000	.000	.000	.000
1	.990	.829	.549	.167	.035	.005	.000	.000	.000	.000	.000	.000	.000
2	1.000	.964	.816	.398	.127	.027	.004	.000	.000	.000	.000	.000	.000
3	1.000	.995	.944	.648	.297	.091	.018	.002	.000	.000	.000	.000	.000
4	1.000	.999	.987	.838	.515	.217	.059	.009	.001	.000	.000	.000	.000
5	1.000	1.000	.998	.939	.722	.403	.151	.034	.004	.000	.000	.000	.000
6	1.000	1.000	1.000	.982	.869	.610	.304	.095	.015	.001	.000	.000	.000
7	1.000	1.000	1.000	.996	.950	.787	.500	.213	.050	.004	.000	.000	.000
8	1.000	1.000	1.000	.999	.985	.905	.696	.390	.131	.018	.000	.000	.000
9	1.000	1.000	1.000	1.000	.996	.966	.849	.597	.278	.061	.002	.000	.000
10	1.000	1.000	1.000	1.000	.999	.991	.941	.783	.485	.164	.013	.001	.000
11	1.000	1.000	1.000	1.000	1.000	.998	.982	.909	.703	.352	.056	.005	.000
12	1.000	1.000	1.000	1.000	1.000	1.000	.996	.973	.873	.602	.184	.036	.000
13	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.995	.965	.833	.451	.171	.010
14	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.995	.965	.794	.537	.140

h. $n = 20$

$\begin{smallmatrix} p \\ k \end{smallmatrix}$	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.818	.358	.122	.012	.001	.000	.000	.000	.000	.000	.000	.000	.000
1	.983	.736	.392	.069	.008	.001	.000	.000	.000	.000	.000	.000	.000
2	.999	.925	.677	.206	.035	.004	.000	.000	.000	.000	.000	.000	.000
3	1.000	.984	.867	.411	.107	.016	.001	.000	.000	.000	.000	.000	.000
4	1.000	.997	.957	.630	.238	.051	.006	.000	.000	.000	.000	.000	.000
5	1.000	1.000	.989	.804	.416	.126	.021	.002	.000	.000	.000	.000	.000
6	1.000	1.000	.998	.913	.608	.250	.058	.006	.000	.000	.000	.000	.000
7	1.000	1.000	1.000	.968	.772	.416	.132	.021	.001	.000	.000	.000	.000
8	1.000	1.000	1.000	.990	.887	.596	.252	.057	.005	.000	.000	.000	.000
9	1.000	1.000	1.000	.997	.952	.755	.412	.128	.017	.001	.000	.000	.000
10	1.000	1.000	1.000	.999	.983	.872	.588	.245	.048	.003	.000	.000	.000
11	1.000	1.000	1.000	1.000	.995	.943	.748	.404	.113	.010	.000	.000	.000
12	1.000	1.000	1.000	1.000	.999	.979	.868	.584	.228	.032	.000	.000	.000
13	1.000	1.000	1.000	1.000	1.000	.994	.942	.750	.392	.087	.002	.000	.000
14	1.000	1.000	1.000	1.000	1.000	.998	.979	.874	.584	.196	.011	.000	.000
15	1.000	1.000	1.000	1.000	1.000	1.000	.994	.949	.762	.370	.043	.003	.000
16	1.000	1.000	1.000	1.000	1.000	1.000	.999	.984	.893	.589	.133	.016	.000
17	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.996	.965	.794	.323	.075	.001
18	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.999	.992	.931	.608	.264	.017
19	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.999	.988	.878	.642	.182

(continued)

TABLE XIV Critical Values of T_L and T_U for the Wilcoxon Rank Sum Test: Independent Samples

Test statistic is the rank sum associated with the smaller sample (if equal sample sizes, either rank sum can be used).

a. $\alpha = .025$ one-tailed; $\alpha = .05$ two-tailed

$n_2 \backslash n_1$	3		4		5		6		7		8		9		10	
	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U
3	5	16	6	18	6	21	7	23	7	26	8	28	8	31	9	33
4	6	18	11	25	12	28	12	32	13	35	14	38	15	41	16	44
5	6	21	12	28	18	37	19	41	20	45	21	49	22	53	24	56
6	7	23	12	32	19	41	26	52	28	56	29	61	31	65	32	70
7	7	26	13	35	20	45	28	56	37	68	39	73	41	78	43	83
8	8	28	14	38	21	49	29	61	39	73	49	87	51	93	54	98
9	8	31	15	41	22	53	31	65	41	78	51	93	63	108	66	114
10	9	33	16	44	24	56	32	70	43	83	54	98	66	114	79	131

b. $\alpha = .05$ one-tailed; $\alpha = .10$ two-tailed

$n_2 \backslash n_1$	3		4		5		6		7		8		9		10	
	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U
3	6	15	7	17	7	20	8	22	9	24	9	27	10	29	11	31
4	7	17	12	24	13	27	14	30	15	33	16	36	17	39	18	42
5	7	20	13	27	19	36	20	40	22	43	24	46	25	50	26	54
6	8	22	14	30	20	40	28	50	30	54	32	58	33	63	35	67
7	9	24	15	33	22	43	30	54	39	66	41	71	43	76	46	80
8	9	27	16	36	24	46	32	58	41	71	52	84	54	90	57	95
9	10	29	17	39	25	50	33	63	43	76	54	90	66	105	69	111
10	11	31	18	42	26	54	35	67	46	80	57	95	69	111	83	127

Source: From F. Wilcoxon, and R. A. Wilcox. "Some Rapid Approximate Statistical Procedures," 1964, 20–23. Courtesy of Lederle Laboratories Division of American Cyanamid Company, Madison, NJ.